Korognai SMO

Reference Implementation — Full Documentation

A single-file, browser-based demonstration of a Service Management & Orchestration (SMO) platform for O-RAN networks — every tab, every term, every scenario explained.

Prepared July 2026 · Synthetic demo, no live network or vendor affiliation

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1. Purpose & How to Use This Document

This document explains, in full detail, every tab, metric, term, and simulated scenario inside the Korognai SMO reference demo. It is written for two audiences at once: someone with deep telecom background who wants to verify the technical grounding, and someone with no telecom background who wants to understand what they're looking at and why it matters.

> **What this demo is:** a single, self-contained HTML file that runs entirely in a web browser, with no backend server and no live network behind it. Every number, alarm, and event on screen is generated live, in-browser, by JavaScript — it is a working simulation of how a real SMO platform behaves, not a recording or a mockup of static screenshots.

> **What this demo is not:** it is not connected to any real radio network, it does not use any vendor's proprietary software or confidential material, and none of the named companies, products, or trademarks of any commercial SMO vendor appear in it. Where real named products were researched to ground the demo's realism, the demo itself uses only generic, industry-standard terminology drawn from public O-RAN Alliance, 3GPP, TM Forum, NIST, and CIS material.

The rest of this document walks through the demo in the same order the tabs appear, left to right. Section 2 defines every acronym and concept used anywhere in the demo, so later sections can reference them without re-explaining each time. If a term appears in `this style`, it is defined in Section 2.

2. Core Concepts & Glossary

2.1 The physical radio network, in plain terms

A mobile network's radio side used to be one box at each cell tower. Modern networks split that single box into three functional pieces so they can be built more flexibly and partly run in the cloud:

Term	Full name	Plain-English meaning
`RU`	Radio Unit	The physical hardware at the antenna. Converts digital signals into actual radio waves and back. Stays at the tower.
`DU`	Distributed Unit	Handles time-critical, millisecond-scale processing. Usually located physically close to the tower because timing is so strict.
`CU`	Centralized Unit	Handles less time-critical processing and can be centralized to serve many towers at once. Split into CU-CP (control) and CU-UP (user data).
`Core`	Core Network	The wired part of the network the radio side connects into — routes traffic to the internet or other networks.

Transport segments

The links between these pieces have their own names, because their timing and bandwidth requirements are all different:

Segment	Connects	Timing sensitivity
Fronthaul	RU ↔ DU	Extremely high — microseconds
Midhaul	DU ↔ CU	High — single-digit milliseconds
Backhaul	CU ↔ Core	Moderate

2.2 The management & intelligence layers

Term	Full name	Plain-English meaning
`SMO`	Service Management & Orchestration	The top-level management "brain" that oversees the whole radio network from above — monitoring, configuring, securing, and automating it.
`Non-RT RIC`	Non-Real-Time RAN Intelligent Controller	The slower (greater than 1 second, often minutes+) intelligence layer that lives inside the SMO. Hosts rApps. Handles longer-term optimization and AI/ML model training.
`Near-RT RIC`	Near-Real-Time RAN Intelligent Controller	A separate platform, physically closer to the radio, making real-time decisions on a 10-millisecond-to-1-second timescale — fast enough that a human, or even the Non-RT RIC, couldn't react in time. Hosts xApps.
`rApp`	(RIC app)	An automated behavior/application hosted on the Non-RT RIC — e.g. energy saving, mobility tuning. Slower timescale, more sophisticated analysis.
`xApp`	(RIC app)	An automated real-time behavior hosted on the Near-RT RIC — e.g. dynamic spectrum sharing. Fast timescale, simpler/quicker decisions.

2.3 Interfaces — how the layers talk to each other

O-RAN Alliance defines a set of open, standardized interfaces so that equipment from different vendors can interoperate. This is the single most important reference table for understanding how any part of the demo connects to any other.

Interface	Connects	Purpose
`Open Fronthaul (O-FH)`	RU ↔ DU	Standardized fronthaul, based on a specific processing "cut point" called functional split 7.2x — enables multi-vendor RU/DU interoperability.
`F1`	DU ↔ CU	Midhaul interface, split into F1-C (control) and F1-U (user plane).
`E1`	CU-CP ↔ CU-UP	Separates control-plane and user-plane processing within the CU.
`E2`	Near-RT RIC ↔ RU/DU/CU	Carries real-time RAN data up, and real-time control actions back down. The xApp control channel.
`A1`	Non-RT RIC (SMO) ↔ Near-RT RIC	Carries AI/ML-informed policy guidance and enrichment data down from the slower layer to the faster one.
`O1`	SMO ↔ all managed RAN elements	FCAPS — fault, configuration, accounting, performance, and security management.
`O2`	SMO ↔ O-Cloud infrastructure	Cloud infrastructure and workload orchestration/lifecycle management.
`R1`	rApps ↔ Non-RT RIC	How an rApp registers itself, declares what data it needs, and receives that data — still being finalized

		in O-RAN standards work as of this writing.

2.4 FCAPS — the five pillars of network monitoring

Letter	Stands for	Plain-English meaning
F	Fault Management	Tracking things that have gone wrong (alarms), so they can be acknowledged and fixed.
C	Configuration Management	A record of every setting on the network and every change made to it.
A	Accounting Management	Tracking usage and resource consumption (often for billing or capacity planning).
P	Performance Management	Tracking how well the network is actually running — speed, reliability, capacity.
S	Security Management	Access control, audit trails, and protecting the network from misuse.

2.5 Radio & performance terms used throughout the demo

Term	Plain-English meaning
PRB (Physical Resource Block)	The basic unit of radio capacity a cell tower has to allocate to users. "PRB utilization" is how much of that capacity is currently in use — like asking what percentage of a highway's lanes are occupied. High PRB utilization means the tower is busy or crowded.
Handover (HO)	What happens when a phone moves and switches from one cell tower's coverage to another's, automatically, without dropping the call. A "handover failure" means that switch didn't complete cleanly — often experienced as a dropped call or a brief lag.
RSRP	Reference Signal Received Power — a measurement of how strong a phone's signal from a given tower actually is.
Drop Call Rate	The percentage of calls that get disconnected unexpectedly mid-call.
Call Setup Success Rate	The percentage of attempts to start a call (or data session) that succeed on the first try.
Network Availability	The percentage of time the network is actually up and reachable. This is why it's capped at 100% in the demo — it is a percentage of uptime, and no system can be "available" more than all of the time.

2.6 Automation concepts

Term	Plain-English meaning
Closed loop	A fully automated cycle: observe a condition → decide what to do → act → verify the result — with no human step in between. Contrast with "open loop," where the system only recommends an action for a human to approve.
SON (Self-Organizing Networks)	The long-established industry term for exactly this kind of automated, closed-loop network optimization — rApps are essentially SON formalized under open, standardized interfaces.
Conflict management	What happens when two automated behaviors try to change the same thing at the same time in different directions — e.g. one rApp wants to raise a tower's power while another wants to lower it. See Section 11 for the three resolution strategies used in this demo.
Intent-based management	Telling the network *what* outcome you want (e.g. "guarantee low latency for this customer") rather than *how* to achieve it — the system translates that intent into the actual technical settings.
Agentic AI	AI that can reason, plan, and take multi-step action toward a goal, rather than just following a fixed rule. This is an early-stage, industry-wide direction, not something fully mature anywhere yet — the demo's "Copilot narration" feature is a deliberately modest, honest example: it explains decisions in plain language rather than claiming full autonomous reasoning.

2.7 Network slicing terms

Term	Plain-English meaning
Network slice	A virtual, separate network carved out of one shared physical network, with its own guarantees — e.g. a guaranteed-speed lane for one enterprise customer, running alongside everyone else's regular traffic on the same towers.
NSSI	Network Slice Subnet Instance — the specific piece of a slice that lives in the radio access network, as opposed to the core network or transport.
S-NSSAI	The identifier a network uses internally to tell which slice a piece of traffic belongs to.
5QI	A standardized code that tells the network what quality-of-service treatment (delay tolerance, priority, reliability) a particular type of traffic needs.
SLA	Service Level Agreement — a measurable, contractual promise about network quality (e.g. "99.9% uptime," "under 10ms latency").

2.8 Cloud & infrastructure terms

Term

	Plain-English meaning
O-Cloud	The generic O-RAN term for the cloud computing infrastructure that hosts virtualized network functions (as opposed to dedicated, single-purpose hardware).
CNF	Cloud-native Network Function — network software packaged to run in containers (like Kubernetes), the modern alternative to dedicated hardware boxes.
vDU / vCU-CP / vCU-UP	Virtualized versions of the DU and CU, running as software on standard servers instead of dedicated hardware.
NF Lifecycle (LCM)	The full set of operations a network function goes through: instantiate (stand up new), heal (auto-recover from a crash), scale (add/remove capacity), upgrade (new software, no downtime), and terminate (clean shutdown).
O2-DMS	O2 Deployment Management Service — the specific interface the SMO uses to tell the cloud infrastructure what software to actually run.
PnP (Plug-and-Play)	New hardware announces itself to the management system and gets configured automatically, with no manual setup step.

2.9 Security & trust terms

Term	Plain-English meaning
Zero-trust (NIST SP 800-207)	A security model where nothing is trusted automatically — every request is checked every time, even from inside the network perimeter. NIST is the U.S. federal standards body that published this model.
CIS Benchmarks	An independent, published checklist of secure configuration settings for common software — used to verify a system is actually hardened, not just assumed to be.
SBOM	Software Bill of Materials — a complete, itemized list of every component inside a piece of software, similar to an ingredients label on food.
Supply-chain integrity	Proof that the software running today is exactly what was built and approved, with nothing tampered with along the way — usually verified through digital signatures.

2.10 Business & commercial terms

Term	Plain-English meaning
TCO	Total Cost of Ownership — the full lifecycle cost of a solution (purchase, energy, labor, integration, maintenance), not just the sticker price.
OPEX	Operating Expenses — the ongoing cost of running something (power, staff, maintenance), as distinct from one-time capital purchase cost.
MTTR / time-to-detect-and-resolve	How long, on average, between a problem starting and it actually being fixed — whether by automation or by a person.

3. Overview Tab

This is the landing page and is deliberately written for a non-technical reader — the goal is that a VP or an operations executive can understand what the platform does within about thirty seconds of opening it, before any of the deeper tabs are explored.

3.1 "In plain terms" callout

A single paragraph at the very top explains the whole platform as "the control room that runs a modern mobile network" — spotting problems before customers notice, fixing many automatically, cutting energy costs, and giving operations staff one screen instead of a dozen disconnected tools. This framing deliberately avoids all acronyms.

3.2 Executive KPI row

KPI	What it measures	Where the number comes from
Issues Resolved Automatically	How many times an rApp actually fixed something today with no human paged	A live counter, incremented every time a real rApp action fires in the Closed Loop simulation
Est. Annual OPEX Saved	Illustrative estimate of savings from energy management and reduced manual labor	A fixed illustrative figure — clearly not a live measurement, since a browser demo has no real cost data to draw from
Avg. Time to Detect & Resolve	How quickly, on average, an issue goes from appearing to being fixed	A fixed illustrative figure, framed against a "vs. hours, manual process" baseline for contrast

3.3 Operational KPI row

Active Alarms and Network Availability link through to the FCAPS Monitor tab; rApps in Closed Loop links through to the Closed Loop tab. All three are clickable, with a hover-highlight affordance, so the Overview page doubles as a navigation hub, not just a summary.

3.4 The cluster grid

Every function in the platform is grouped into eight clusters spanning its full functional scope — RAN architecture, SMO/interfaces, FCAPS assurance, cloud & automation, Non-RT RIC intelligence, Near-RT RIC, slicing/conflict, and security. Each cluster card leads with a plain-English business-value line before the technical description, and is directly clickable through to its corresponding tab.

Cluster	Links to tab
RAN Architecture: RU/DU/CU & Interfaces	RAN Architecture
SMO Architecture & Interfaces	Data & Openness
FCAPS & Assurance	FCAPS Monitor
Cloud & Automation	Cloud & NF Lifecycle
Non-RT RIC: rApps, AI/ML, Agentic	Closed Loop
Near-RT RIC: xApps & Real-Time Control	Near-RT RIC
Slicing, Conflict & Intent	Slice Lifecycle
Security, Integration & Support	Security & Trust

4. RAN Architecture Tab

This tab exists to answer, visually, one of the most commonly asked architecture questions in this space: "*How do SMO, the Non-RT RIC, and the Near-RT RIC divide responsibility, and what actually connects them?*"

4.1 The diagram

Three layers, stacked top to bottom:

- **Top: SMO**, hosting the Non-RT RIC (rApps) — connects to everything below via O1 (FCAPS), O2 (to cloud infrastructure), and A1 (policy, downward only)
- **Middle: Near-RT RIC**, hosting xApps — connects to the radio elements via E2
- **Bottom: the radio chain itself** — RU, connected via Open Fronthaul to DU, connected via F1 to CU-CP/CU-UP (themselves connected via E1), connected via backhaul to the Core network

4.2 The hover-highlight interaction

Every interface label in the diagram, and every row in the interface reference table beneath it, is wired to a shared highlight function. Hovering an interface anywhere — in the diagram or the table — highlights its counterpart in both places simultaneously. This was specifically built (and then corrected, after an initial version only partially worked for the combined O1/O2/A1 line) so that a viewer never has to mentally map a label in one place to a row somewhere else; pointing at "E2" lights up "E2" everywhere it appears.

> Every box in the diagram (RU, DU, CU-CP/CU-UP, Core, SMO, Near-RT RIC) also carries its own hover tooltip with a plain-English explanation, independent of the interface highlighting.

4.3 Interface reference table

See Section 2.3 above for the full, identical table used in this tab.

5. FCAPS Monitor Tab

This tab is the piece that was added specifically to cover core network monitoring, since the platform's other tabs focus heavily on automation and intelligence rather than the underlying day-to-day monitoring discipline every SMO is fundamentally built on.

5.1 Fault Management — active alarms

A live list of alarms, each with a severity level, a source (which cell or network function raised it), a description, a time-since-raised counter, and an acknowledge action.

Severity	Meaning
Critical	Something has completely stopped working — e.g. a site outage
Major	Significant degradation that needs attention soon — e.g. sustained high load
Minor	A condition worth tracking but not yet service-impacting — e.g. rising CPU use
Warning	An early trend worth watching — e.g. a slowly rising failure rate

Acknowledging an alarm marks it as reviewed (it fades slightly and shows an "acknowledged" pill) without deleting it — reflecting how real fault management preserves a record even after a human has taken ownership of an issue.

5.2 Performance Management — live KPIs

Four KPIs tick continuously, each on a bounded random walk (see Section 5.2.1) so they feel alive without ever drifting to an unrealistic value:

KPI	Bounds	Why bounded this way
RAN Availability	99.5% – 100%	Availability is a percentage of uptime and can never exceed 100% — this was a bug in an earlier version of the demo, fixed after being reported, since an unbounded random walk could previously drift the number above 100%, which is physically meaningless.
Call Setup Success Rate	95% – 100%	Reflects a healthy but not impossibly perfect network
Avg. DL Throughput / cell	120 – 260 Mbps	A realistic range for a modern 5G cell under mixed load
Drop Call Rate	0.05% – 2.5%	Kept low, since a healthy network should rarely drop calls, but never exactly zero

5.2.1 Why a "bounded random walk"?

Rather than picking a new random number every tick (which would look jumpy and unrealistic), each KPI takes a small random step from its current value, then that step is clamped to the KPI's realistic min/max range. This produces a value that drifts smoothly, the way a real network's performance actually does, while never producing an impossible reading.

5.3 Configuration Management — recent changes

A change log showing, for each recent configuration change: the timestamp, which network entity was affected, which parameter changed, the old and new values, and — importantly — *who or what made the change*. Several entries are attributed to specific rApps (e.g. "Energy Saving Management" changing a power parameter), directly illustrating that automated systems, not only human engineers, are configuration change sources that need to be tracked and auditable.

5.4 Trace sessions

A "trace" follows one specific call or device through the network in detail, used for deep troubleshooting when a specific customer's experience needs to be understood precisely (as opposed to aggregate KPIs). The tab allows starting a new trace against a randomly selected cell, which transitions from "running" to "complete" after a short delay, mirroring the real workflow of requesting and waiting on a diagnostic trace.

5.5 Topology & inventory summary

Three rollup counts: number of sites, number of radio network functions (RU/DU/CU instances), and number of distinct RAN vendors integrated — representing the "single source of truth" inventory that every other automation function in the platform ultimately depends on being accurate.

6. Closed Loop Tab & Scenario Reasoning

This is the centerpiece of the demo: a live visualization of the Non-RT RIC actually detecting problems and fixing them automatically, with real O-RAN SON-style algorithms behind each fix rather than placeholder logic.

6.1 The cell ring

Twelve cells, drawn from three different simulated RAN vendors, arranged in a ring around a central "Non-RT RIC" hub. Each cell is colored by its current state (green = nominal, amber = congested, red = alarm, cyan = an rApp is actively remediating it), with an additional colored outer ring indicating which of the three vendors that cell belongs to — making the "multi-vendor" aspect visible at a glance rather than only implied by a label. Hovering any cell shows its ID, vendor, state, live PRB/handover metrics, and the last rApp action taken on it.

6.2 The two view modes: sample vs. fleet scale

A toggle switches between the 12-cell interactive ring and a 10,000-cell "fleet view," which deliberately uses a completely different visualization technique — a canvas-rendered density heatmap with rollup statistics and a ranked exceptions list, rather than 10,000 individual dots. This exists to make a specific engineering point: individual-object visualizations do not scale to real network sizes, and the correct pattern at scale is **aggregation with drill-down** — rollup numbers and a ranked worklist of exceptions, with the detailed 12-cell view reserved for whichever site an operator actually needs to drill into.

6.3 The four trigger scenarios — detailed reasoning

Each button injects a specific, realistic fault or performance condition onto a randomly chosen cell, and is color-coded by severity (amber for a performance condition, cyan for a neutral condition, red for a fault, neutral gray for reset). Clicking a button produces three layers of confirmation feedback: the button itself flashes, a toast notification appears describing exactly what happened and to which cell, and a red shockwave ring visibly pulses outward from the affected cell in the diagram — added specifically because an earlier version of the demo gave no clear confirmation that a click had actually done anything.

Scenario: Simulate load spike Amber

What it does: Sets a randomly chosen cell's PRB utilization to 88% and marks its state as "congested."

Real-world meaning: This simulates a sudden surge in demand on one tower — for example, a large public event, a stadium emptying out, or a viral moment causing many nearby phones to demand data simultaneously.

What responds: If the Mobility Load Balancing rApp is enabled, it detects PRB utilization above its configured threshold (75%) and automatically offloads a portion of idle-mode users to neighboring, less-busy cells, stepping utilization back down.

Why this matters operationally: Without automated load balancing, a congested cell degrades every user's experience on it simultaneously (slower speeds, more dropped connections) until an engineer manually intervenes or the surge naturally subsides — which could be hours.

Scenario: Simulate sleeping cell Cyan

What it does: Sets a randomly chosen cell's PRB utilization to 8%, keeping its state "nominal" (this is not a fault — it is a condition, hence the neutral cyan color rather than a warning color).

Real-world meaning: This simulates a cell that is sitting mostly idle — for instance, a tower in a business district at 3am, or a rural cell with very light traffic overnight.

What responds: If the Energy Saving Management rApp is enabled, it detects sustained low utilization below its configured threshold (20%) and applies a micro cell-sleep mode, reducing the cell's power draw without materially affecting the few users still connected to it.

Why this matters operationally: Radio equipment is one of an operator's largest ongoing electricity costs. A tower running at full power to serve almost no users overnight is pure waste — this is the single most quantifiable, provable value case for rApp automation in the whole industry, and is deliberately included as its own dedicated scenario for that reason.

Scenario: Simulate cell outage Red

What it does: Sets a randomly chosen cell's state to "alarm" and its PRB utilization to 95% (representing neighboring cells absorbing displaced traffic).

Real-world meaning: This simulates a hard site failure — a power outage at the tower, a hardware fault, a fiber cut to that site, or similar. The site has effectively gone dark, and its coverage area now has a hole in it.

What responds: If the Cell Outage Compensation rApp is enabled, neighboring cells automatically adjust their antenna tilt to widen coverage and partially fill the gap left by the failed site, while the fault is separately raised in Fault Management for a human to actually repair the underlying hardware or power issue.

Why this matters operationally: A total outage is the most severe category of network fault — customers in the affected area can lose service entirely. Automated coverage compensation doesn't fix the root cause, but it meaningfully reduces the customer-facing impact in the window between the failure occurring and a technician physically resolving it, which is exactly why it is modeled as a distinct, separate capability from the underlying fault alarm itself.

Scenario: Reset to nominal Neutral

What it does: Regenerates all 12 cells back to random, healthy baseline values, clearing any injected condition.

Why it exists: Purely a demo-usability feature — it lets a presenter cleanly return to a calm starting state between scenarios rather than accumulating multiple simultaneous faults, which is why it is styled as a neutral "ghost" button rather than color-coded like the three fault-injecting scenarios.

6.4 Copilot narration (plain-language vs. technical log toggle)

The R1 interface event feed can be switched between a raw technical log (e.g. "[CELL-107] Mobility Load Balancing: Offloading 10% of idle-mode UEs to neighbor cells") and a plain-language narration of the same event (e.g. "Cell 107 was getting overloaded, so the network nudged some phones to a nearby, less-busy tower instead"). This exists to demonstrate a real industry principle: agentic and autonomous systems are expected to explain their own decisions in a form a non-specialist can audit, not operate as an unexplained black box.

7. Near-RT RIC Tab

This tab exists specifically to close a gap: without it, the whole demo would only ever show the *slow* half of RAN intelligence (the Non-RT RIC), when a complete picture requires showing both halves and, critically, the actual timescale difference between them.

7.1 The explicit contrast

A callout at the top states plainly: everything on the Closed Loop tab is the Non-RT RIC, running on a greater-than-1-second (often minutes+) timescale. This tab is the Near-RT RIC, running on a 10-millisecond-to-1-second timescale over the E2 interface — control decisions simply too fast for the slower rApp logic to make in time. The Non-RT RIC still steers the Near-RT RIC, but only via slower A1 policy guidance, not by micromanaging every millisecond itself.

7.2 Making the timescale difference visible, not just stated

A live "E2 Control Actions / sec" counter ticks far faster than the rApp action counter on the Closed Loop tab — the demo runs an E2 control tick every 150 milliseconds, versus the Non-RT RIC's roughly one action every few seconds. This was a deliberate design choice: simply writing "10ms-1s vs. greater-than-1s" in text is easy to skim past, while watching one counter visibly race ahead of another makes the distinction impossible to miss.

7.3 The xApp catalog

xApp	What it does
Dynamic Spectrum Sharing	Real-time arbitration of shared spectrum between 4G and 5G carriers, slot by slot
Real-Time Load Balancing	Millisecond-scale traffic steering between cells based on instantaneous radio conditions
Interference Mitigation	Detects and reacts to interference within the same control interval it is observed in
QoS-Aware Scheduling Assist	Adjusts scheduler weighting in real time to protect latency-sensitive traffic flows

Each is individually toggleable, using the same interaction pattern as the rApp Catalog tab, for consistency across the two "app catalog" experiences in the demo.

8. rApp Catalog Tab

Six rApps are modeled with genuine, runnable reference logic (not just descriptive text) — each one is real O-RAN SON-style automation, grounded in generic industry requirement categories rather than any single vendor's proprietary product.

rApp	Category	What it actually does (live logic)
Energy Saving Management (ESM)	AI/ML forecast	If a cell's PRB utilization stays below 20% for a sustained period, applies a micro cell-sleep mode, reducing power draw.
Automatic Neighbor Relation (ANR)	Configuration Management	Models the ongoing discovery and cleanup of which cell towers are adjacent to each other, across vendor boundaries.
Mobility Robustness Optimization (MRO)	Performance Management	If a cell's handover failure rate exceeds 3.0%, increases handover hysteresis by 1dB, smoothing out the transition and reducing failures.
Mobility Load Balancing (MLB)	Performance Management	If a cell's PRB utilization exceeds 75%, offloads roughly 10% of idle-mode users to a neighboring, less-busy cell.
Cell Outage Compensation (COC)	Fault Management / Self-Healing	If a cell enters alarm state, adjusts a neighboring cell's antenna tilt to widen coverage and partially fill the resulting gap.
PCI Reuse Optimization (PCI)	Configuration Management	Models detection and correction of Physical Cell Identity collisions between nearby towers.

Each card also shows which R1-style data services it subscribes to (e.g. cell metrics, PRB utilization) — reflecting the real O-RAN principle that an rApp declares its data dependencies explicitly through the R1 interface rather than reaching into arbitrary network data directly.

9. Import rApp Tab

This tab demonstrates the "bring your own rApp" capability that is central to O-RAN's open-ecosystem vision — the idea that an operator should be able to onboard a third-party or in-house-developed rApp onto a common platform, not be limited to whatever one vendor pre-built.

9.1 The descriptor schema

Modeled on real Non-RT RIC rApp Manager conventions, a valid descriptor must declare: `rAppId`, `name`, `version`, `requiredServices`, `exposedServices`, `policyType`, and a `logic` object naming an algorithm and its parameters.

9.2 Honest handling of unrecognized logic

If the submitted descriptor's `logic.algorithm` matches one of the four algorithms the demo actually implements (energy saving, mobility robustness, load balancing, or outage compensation), the imported rApp runs genuine logic identical to the built-in versions. If it names anything else, the rApp is honestly onboarded as a **monitoring-only stub** rather than silently pretending to execute logic that doesn't exist — this distinction is deliberate and important: a real platform should never claim to autonomously act on logic it doesn't actually have.

9.3 Why this reflects a real technical nuance

In an actual O-RAN deployment, a package descriptor mostly describes *deployment metadata* — which R1 services an rApp consumes, its resource requirements — while the actual decision algorithm lives in the rApp's own proprietary code. The demo's descriptor format mirrors that real convention, while adding a `logic` block purely so the demo can actually execute something when a recognized algorithm name is submitted.

10. Slice Lifecycle Tab

A slice (see Section 2.7) is not a one-time configuration — it is a living, managed entity across its whole life. This tab demonstrates that full lifecycle rather than only a static "here is a slice" snapshot.

10.1 The four lifecycle stages

Stage	What happens
1. Create	NSSI creation request accepted, S-NSSAI allocated, RAN resource template applied
2. Modify	NSSI modification applied without service disruption — e.g. a PRB partition updated at runtime
3. Monitor & Assure	Continuous SLA monitoring against the slice's committed targets
4. Decommission	NSSI decommission executed, resources released, inventory reconciled

10.2 Business intent translation

Three example business intents (guarantee low latency for an enterprise customer, guarantee sustained throughput for a premium consumer tier, maximize capacity for a stadium event) each translate into concrete technical policy — a specific 5QI value, a target Packet Delay Budget, and a scheduling priority weight — with a live bar chart showing actual performance against the committed SLA target for each active slice.

11. Conflict Log Tab

When multiple rApps act on the same network simultaneously, their recommendations can directly contradict each other — for example, a coverage-optimization rApp raising a cell's transmit power at the same moment an energy-saving rApp is trying to lower it. This tab demonstrates detecting and resolving exactly that kind of conflict.

11.1 The three resolution strategies

Strategy	Behavior
Allow	Both conflicting configurations are applied; the SMO simply monitors the resulting KPI drift rather than blocking either action.
First-Come-First-Served (FCFS)	Whichever rApp's action was applied first is retained; the second, conflicting action is rejected for a cooldown window.
Priority-Based Override	A pre-configured priority ranking decides which rApp's action wins outright, regardless of timing.

11.2 Why this matters

Without conflict management, multiple well-intentioned automated systems can fight each other indefinitely — for example, two rApps repeatedly reversing the same setting back and forth, burning signaling capacity and potentially degrading the very KPI both were trying to improve. This is widely regarded as one of the areas of O-RAN standards still actively maturing, rather than a fully solved problem industry-wide.

12. Data & Openness Tab

12.1 Data coverage matrix

A grid showing, per data domain (Fault Management, raw Performance counters, Configuration Management, Topology & Inventory) and per simulated vendor, whether that data is exposed through an open interface or retained internally by that vendor only. This directly visualizes a common real-world problem: some vendors expose everything through open interfaces, while others keep certain raw data internal to their own tooling — meaningfully limiting how much true cross-vendor automation is actually possible.

12.2 Resilience / decoupling simulator

A button simulates the external data platform becoming completely unreachable. A well-decoupled SMO architecture should continue buffering events locally and keep its closed loops stable through the outage, rather than failing outright — the simulation shows a local buffer counter climbing while disconnected, then a clean catch-up forward-sync once reconnected. This directly feeds the "Decoupled architecture" score on the Openness Scorecard tab (see Section 16).

12.3 Self-serve data routing

A simple form lets a user redirect a data domain to a different destination endpoint, applied immediately with no simulated ticket or maintenance window — illustrating the difference between a genuinely open platform (where the operator controls their own data routing) and a closed one (where every integration change requires vendor professional services).

13. Cloud & NF Lifecycle Tab

13.1 Network function lifecycle actions

Five triggerable actions — Instantiate, Heal, Scale, Upgrade, Terminate — each transitioning through a live "in progress" state before reporting completion, representing the full lifecycle a cloud-native network function goes through, not merely its initial deployment.

13.2 Infrastructure visibility table

A single queryable view of where every network function actually runs — which physical or cloud location is hosting each vDU, vCU-CP, vCU-UP, or Non-RT RIC host instance, and its current status. This represents the "single source of truth" inventory view discussed in Section 5.5, applied specifically to cloud/compute infrastructure rather than radio topology.

14. Zero-Touch Rollout Tab

This tab models a full, thirteen-step commissioning pipeline for bringing a new site online with minimal manual intervention — built entirely using generic, publicly documented O-RAN/3GPP concepts rather than any specific vendor's internal tooling.

14.1 The thirteen steps

1. Onboard software package to the SMO — validate format, signature, and contents against the container registry
2. Auto-discover on-site hardware (RU/DU/CU) via O1 plug-and-play
3. Synchronize the new cloud instance / network element with the SMO inventory
4. Generate site configuration from design inputs (a single engine, all architectures)
5. Create a CNF deployment plan and map configuration values from the generated site config
6. Deploy the CNF to the O-Cloud via O2-DMS, with auto-configuration if PnP is supported
7. Load the site configuration to the network element for commissioning
8. Validate the configuration plan against schema before activation
9. Activate the configuration plan
10. Verify software version and inter-node (e.g. CU-DU) connectivity
11. Pre-launch RF/coverage check
12. Post-launch KPI validation window (holds if any metric is out of bounds)
13. Save the activated configuration as the new baseline; generate a step-by-step audit report

14.2 Why this level of granularity matters

Each step is individually animated and logged, specifically to make the point that "zero-touch" does not mean "no process" — it means a rigorous, auditable, automatically-executed process, with validation gates and rollback points built in, rather than a human simply skipping steps.

15. Security & Trust Tab

15.1 Evidence-based security posture

Four security controls, each individually toggleable, each showing concrete "evidence" text when enabled rather than a bare claim:

Control	Evidence shown when enabled
Zero-trust access (NIST SP 800-207)	Every service call authenticated and authorized — zero implicit-trust paths found
Component hardening (CIS Benchmarks)	Database/web-server images scanned against a specific CIS baseline version, with findings remediated
Supply-chain integrity	SBOM generated, image signatures verified, zero unverified artifacts in the deployment path
Secrets management	Encrypted at rest and in transit, access logged, credentials rotated on a fixed schedule

The deliberate design principle here is "evidence, not assertion" — the tab title itself is "Security & Trust," and the whole point is that a mature platform should be able to show proof of a control working, not merely claim the control exists.

15.2 Supply-chain artifact verification

A button verifies the next artifact in a rotating list (an rApp package, a Helm chart, a container image, an SBOM manifest), logging a signature/checksum verification each time — illustrating ongoing, ordinary-course verification rather than a one-time audit event.

16. Openness Scorecard Tab

Six industry-relevant principles, each scored 0-100, rolling up into a single composite score. Unlike every other metric in the demo, there is no live sensor for "how open is this architecture" — these are inherently qualitative judgment calls, the kind an operator would actually make when scoring a vendor during a procurement evaluation. Rather than inventing a fake precise formula, each principle is instead wired to something genuinely measurable elsewhere in the demo, with the calculation source noted directly under each slider.

Principle	Live data source
Openness & data ownership	Percentage of the Data & Openness coverage matrix marked "exposed"
Decoupled architecture	Resilience demonstrated — rises with each successful reconnect after a simulated data-platform outage
Trustworthy automation	Percentage of currently active rApps running genuine (non-stub) reference logic, versus imported-but-unrecognized stubs
Security by evidence	Percentage of Security & Trust controls currently showing evidence (i.e., toggled on)
Safe autonomy (conflict-aware)	Strength of the currently active conflict-resolution strategy on the Conflict Log tab
Freedom from lock-in	Percentage of the Overview's functional clusters that are actually interactive versus documentation-only

Every slider remains manually adjustable — useful for a "here's how I'd score vendor X" conversation — and a "Recalculate from live demo state" button snaps all six back to what is currently true elsewhere in the demo, so dragging a slider never permanently detaches it from reality unless the user explicitly wants it to.

17. Design Principles

17.1 One file, no backend

Every tab, every simulation, every calculation runs client-side in a single HTML file with embedded CSS and JavaScript. There is no server dependency, which matters specifically for live demonstration reliability — nothing can fail due to network issues, API rate limits, or a backend going down mid-conversation.

17.2 Honesty about synthetic data

Every screen in the demo is explicitly labeled as simulation. This is treated as a hard requirement rather than a nice-to-have: the Import rApp tab, for instance, deliberately labels unrecognized imported logic as a "monitoring-only stub" instead of silently pretending to run behavior that doesn't exist, and the top-level badge on every page states "SIMULATION — SYNTHETIC DATA" without qualification.

17.3 Grounded in public standards only

All terminology, interface names, and behaviors are drawn from public O-RAN Alliance, 3GPP, TM Forum, NIST, and CIS material. No proprietary vendor documentation, confidential material, or trademarked commercial product names are used anywhere in this demo — every automation, interface, and workflow shown reflects generic, industry-standard practice rather than any single company's internal implementation.

17.4 Scale reasoning: aggregation over enumeration

When asked how the 12-cell ring visualization would need to change at real network scale (tens of thousands of cells), the answer was not to simply draw more dots — individual per-object visualizations become both computationally impractical (thousands of live DOM/SVG elements) and cognitively unreadable well before real network scale. The correct, and now implemented, pattern is aggregation with drill-down: a density heatmap using canvas rendering (not per-cell DOM elements), rollup statistics, and a ranked exceptions worklist — with the detailed per-cell ring view explicitly reserved for whatever a user actually needs to drill into, rather than attempting to show everything at once.

17.5 Executive readability alongside technical depth

Following a specific request to make the demo comprehensible to non-technical, senior management readers, plain-language framing was added throughout — an "In plain terms" callout on the Overview page, business-value one-liners preceding every technical description, and a comprehensive hover-tooltip system covering every tab name, KPI title, and jargon abbreviation anywhere in the demo — without removing or diluting the technical depth available to someone who wants it.

17.6 Visible confirmation of every action

Following feedback that triggering a scenario gave no clear confirmation anything had happened, every user-triggered action in the demo now produces layered, immediate feedback — a button-level animation, a toast notification with the specific outcome, and (for the Closed Loop scenarios specifically) a shockwave highlight directly on the affected element — rather than relying on a user to notice a subtle change in a scrolling log or a slowly updating number.

18. Appendix: Quick-Reference Tables

18.1 Full acronym index

Acronym	Full term
SMO	Service Management & Orchestration
RIC	RAN Intelligent Controller
RAN	Radio Access Network
RU / DU / CU	Radio Unit / Distributed Unit / Centralized Unit
O-FH	Open Fronthaul
FCAPS	Fault, Configuration, Accounting, Performance, Security management
PRB	Physical Resource Block
HO	Handover
RSRP	Reference Signal Received Power
SON	Self-Organizing Network
NSSI	Network Slice Subnet Instance
S-NSSAI	Single Network Slice Selection Assistance Information
5QI	5G QoS Identifier
SLA	Service Level Agreement
CNF	Cloud-native Network Function
NF / LCM	Network Function / Lifecycle Management
O2-DMS	O2 Deployment Management Service
PnP	Plug-and-Play
SBOM	Software Bill of Materials
NIST	National Institute of Standards and Technology
CIS	Center for Internet Security
TCO	Total Cost of Ownership
OPEX	Operating Expenses
ESM	Energy Saving Management
ANR	Automatic Neighbor Relation
MRO	Mobility Robustness Optimization
MLB	Mobility Load Balancing
COC	Cell Outage Compensation
PCI	Physical Cell Identity